

- 3 Salmon can be genetically modified (GM) to produce increased quantities of growth hormone, which is a protein. GM salmon modified in this way have a faster growth rate and reach their maximum body mass at a younger age than non-GM salmon.

- (a) Within any population of salmon there is variation in body mass. This is an example of continuous variation.

Explain what is meant by continuous variation **and** how it can be caused.

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- (b) Scientists investigated whether injection of very young non-GM salmon with recombinant growth hormone could cause an increase in the growth rate of the salmon.

The scientists used two groups of non-GM salmon:

- a control group of salmon that were not injected with recombinant growth hormone
- an experimental group of salmon that were injected with $1.0\mu\text{g}$ of recombinant growth hormone at the start of the experiment and once a week for the next six weeks.

The mean body mass of the salmon in the two groups at the start of the experiment was the same (5.3g).

After six weeks, the body mass of every salmon was measured again. The results are summarised in Table 3.1.

Table 3.1

		no injection with recombinant growth hormone	injected with recombinant growth hormone
number of non-GM salmon (n)		28	27
body mass /g	range	6.5–8.6	7.2–12.7
	mean ($\bar{x}$)	7.7	9.4
	standard deviation (s)	0.4	1.1

A student decided that a t -test should be performed on the results shown in Table 3.1.

- (i) Calculate the value of t for the results shown in Table 3.1 using the formula for the t -test:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

Give your answer to **two** decimal places.

Show your working.

$t =$ [3]

- (ii) The critical value at $p = 0.05$ for these data is 2.01.

The student used the results in Table 3.1 and the t -test to conclude that the injections of recombinant growth hormone cause an increase in the growth rate of the non-GM salmon.

Comment on the extent to which the conclusion made by the student can be supported.

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- (iii) Suggest **one** advantage, **other** than cost, of farming GM salmon that produce increased quantities of growth hormone instead of farming non-GM salmon that are injected with recombinant growth hormone each week.

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[Total: 10]